



Genomics of Gene Regulation: Concepts and Techniques

Ana Azambuja

Senior Research Associate

Pathology Department - Boston Children's Hospital



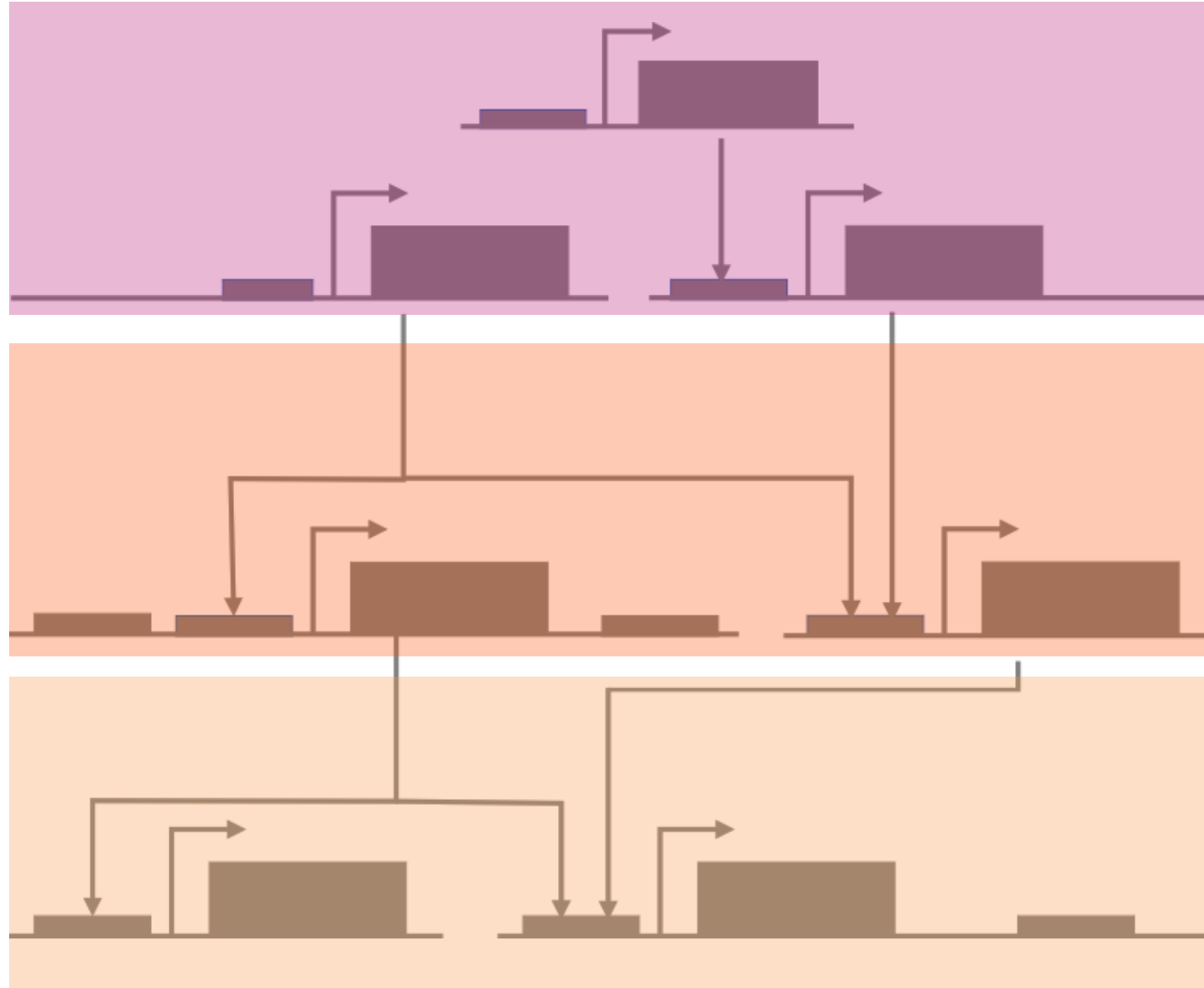
Genomics of Gene Regulation: Concepts and **Techniques**

Chromatin Conformation
Enhancer activation

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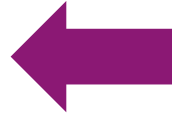
Cellular transitions are controlled by GENE REGULATORY NETWORKS



Cis-regulatory features and how to identify them

DNase-seq

DNase I hypersensitive sites sequencing

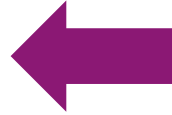


ATACseq

Assay for Transposase-Accessible Chromatin

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Chromatin Immunoprecipitation

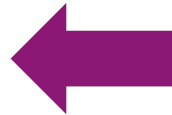


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Cleavage Under Targets & Release Using Nuclease

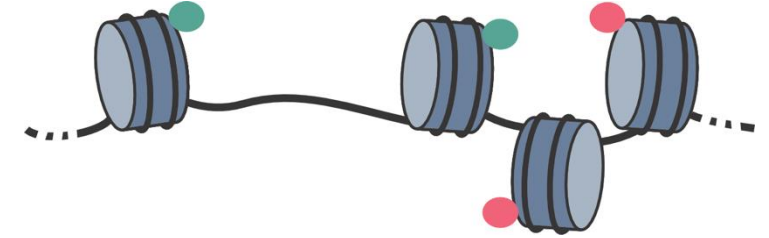
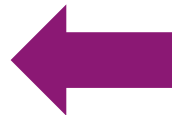
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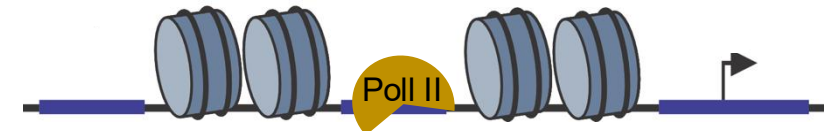
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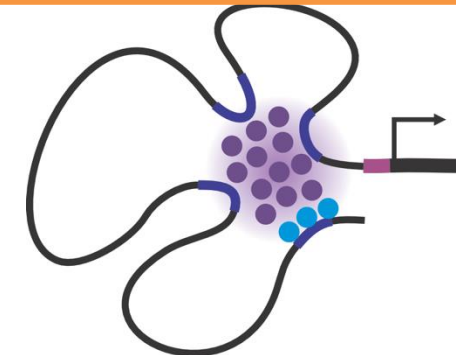
DNA accessibility and active histone marks



Tissue specific transcription factor binding

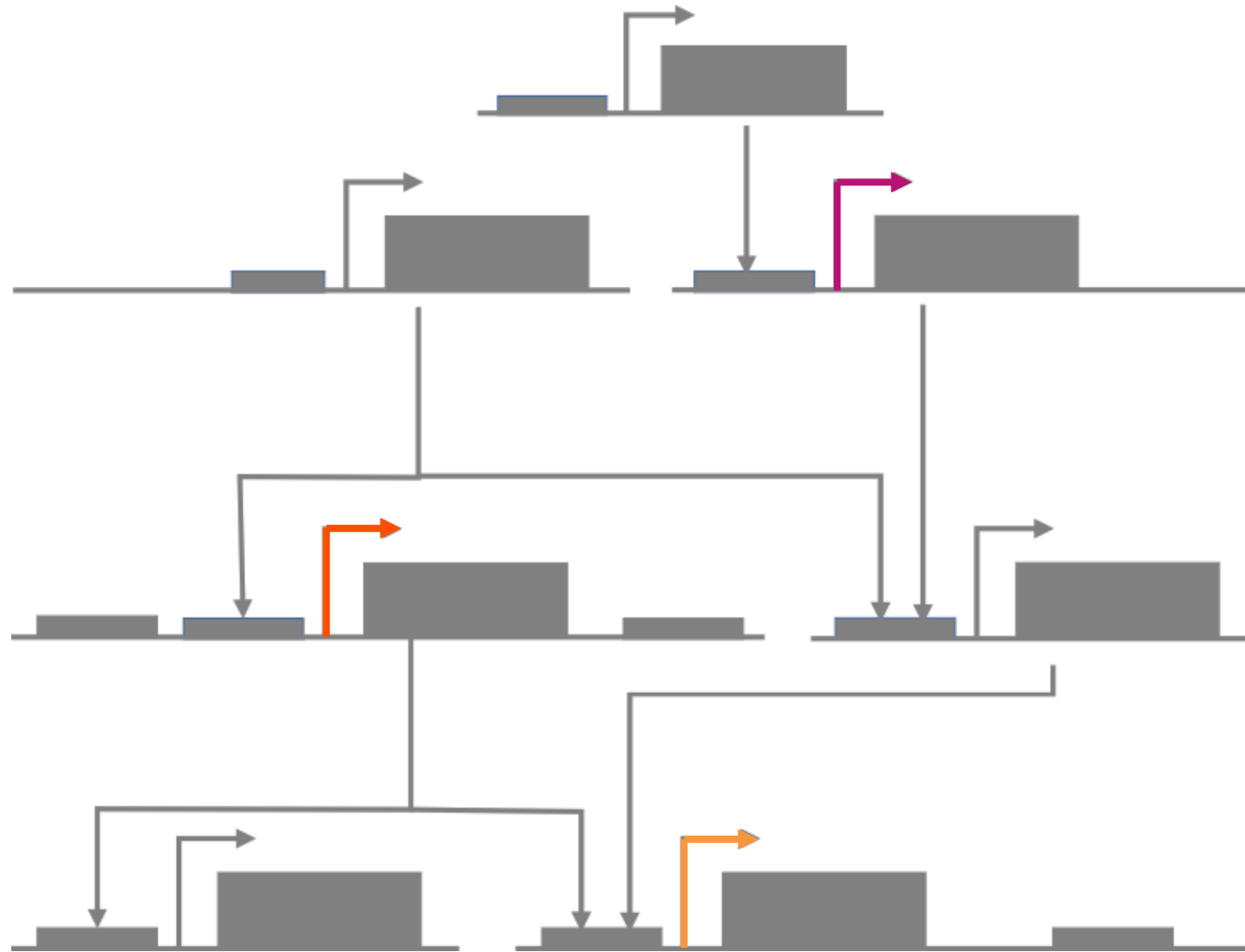


Enhancer activation and transcription



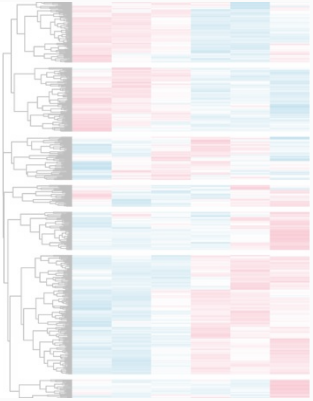
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Topics to be covered - Techniques

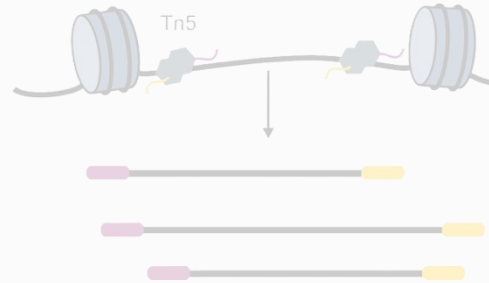
GENE EXPRESSION



RNAseq

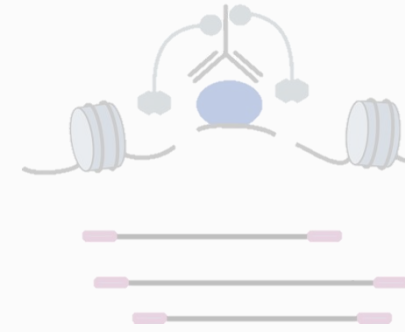
Spatial transcriptomics

EPIGENOMIC REGULATION



ATACseq

DNA-PROTEIN BINDING



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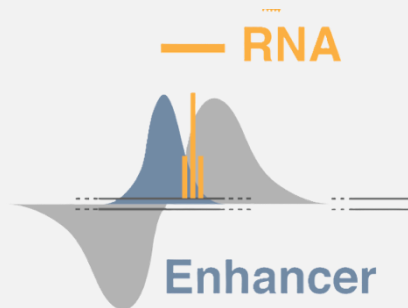
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3D genome organization - HiC



ENHANCER ACTIVITY

eRNA mapping

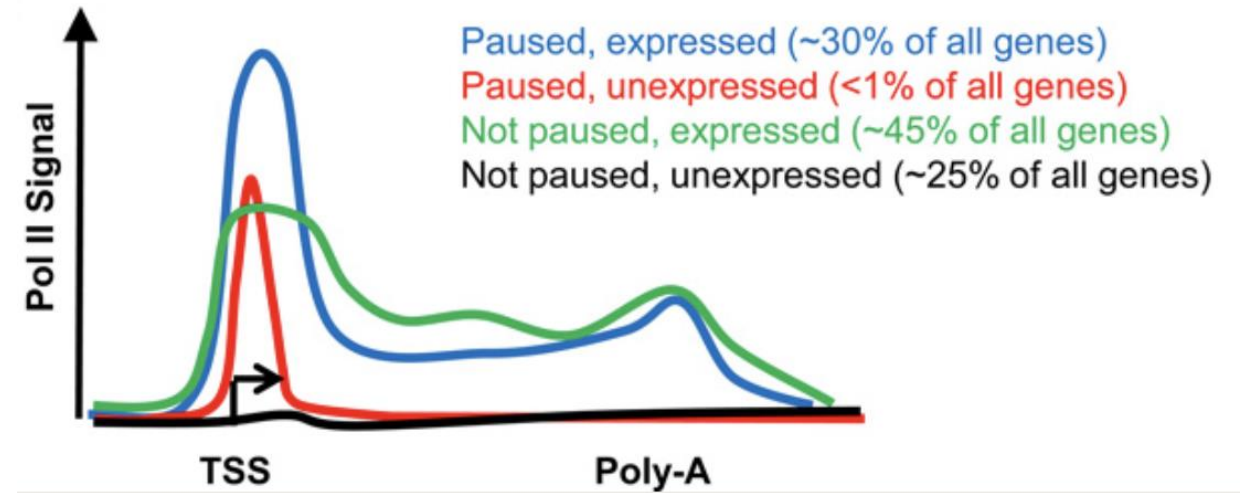
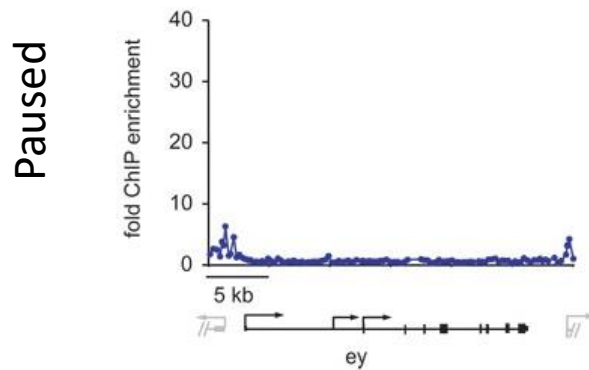
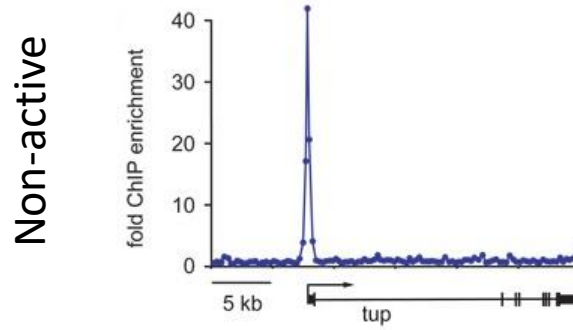
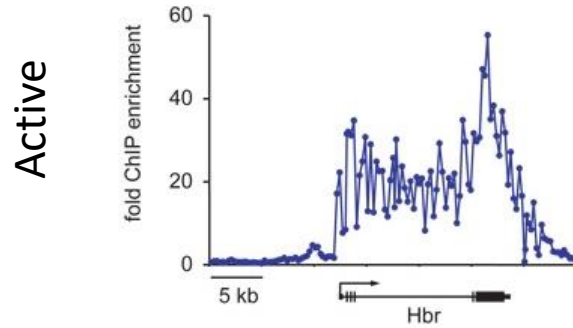


SINGLE-CELL TECHNOLOGIES

Transcriptomics / Multiome



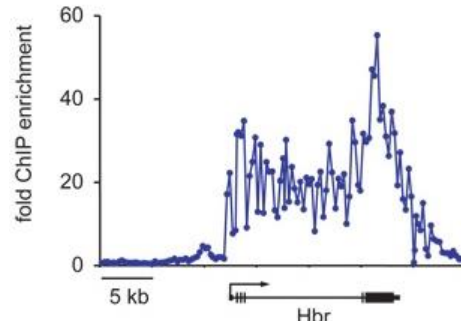
Mapping active transcription



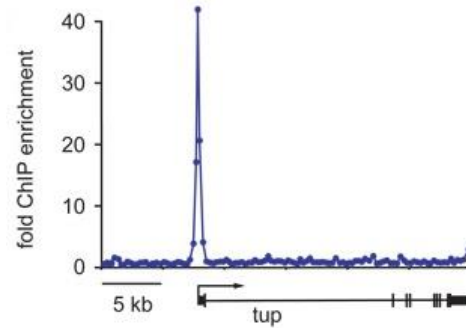
Mapping active transcription

Nascent RNA sequencing (run-on assays)

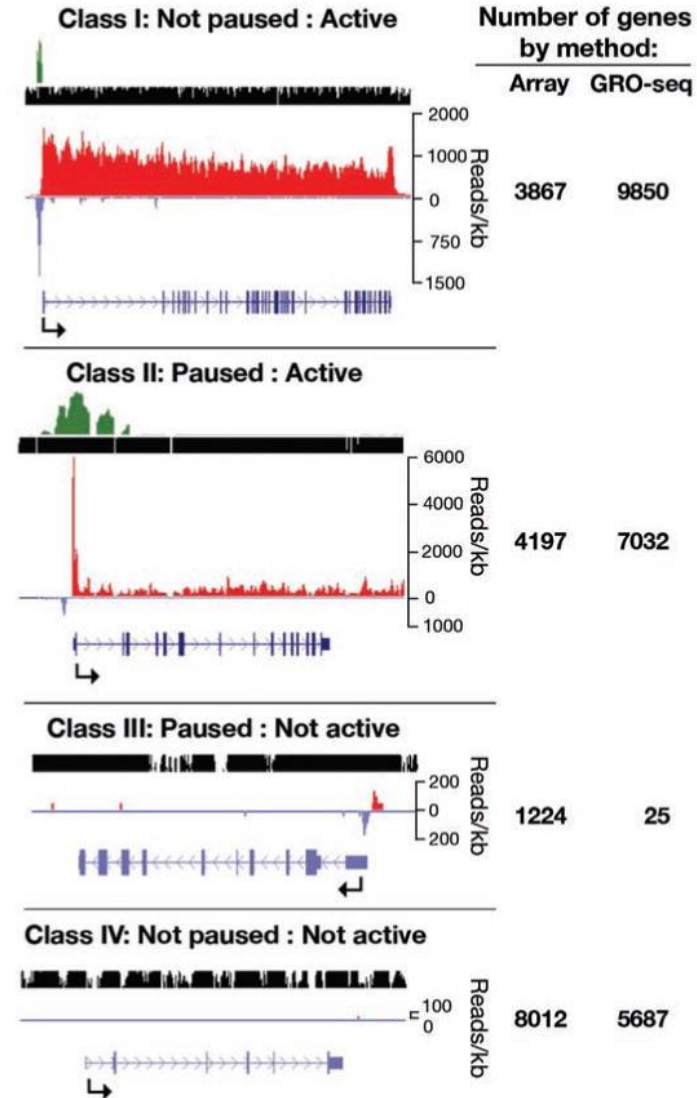
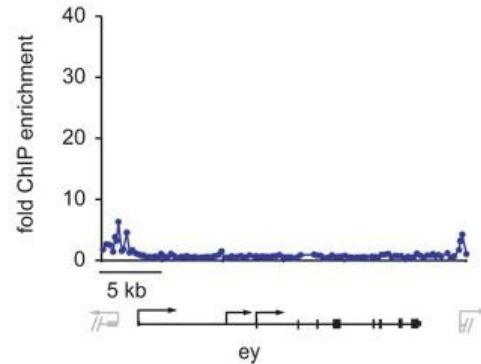
Active



Non-active

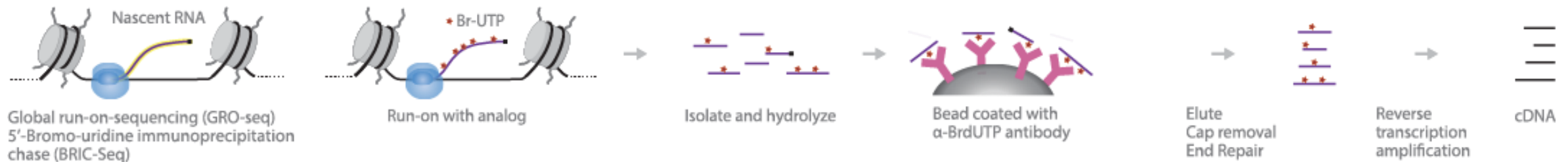


Paused



Mapping active transcription and unstable RNA

GRO-seq - global run-on sequencing



Pros:

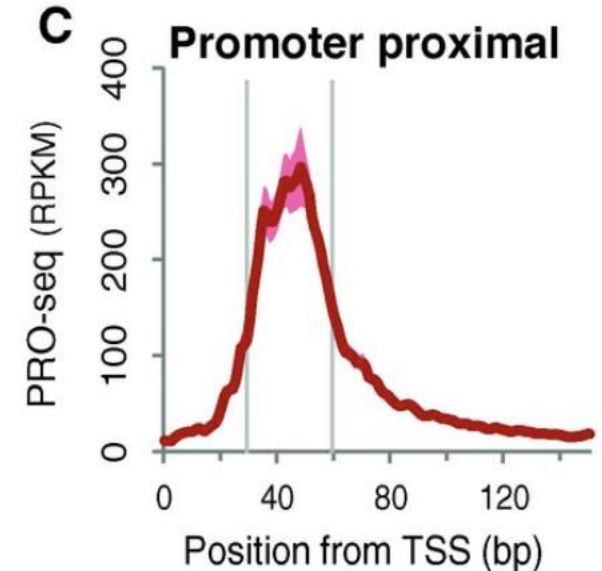
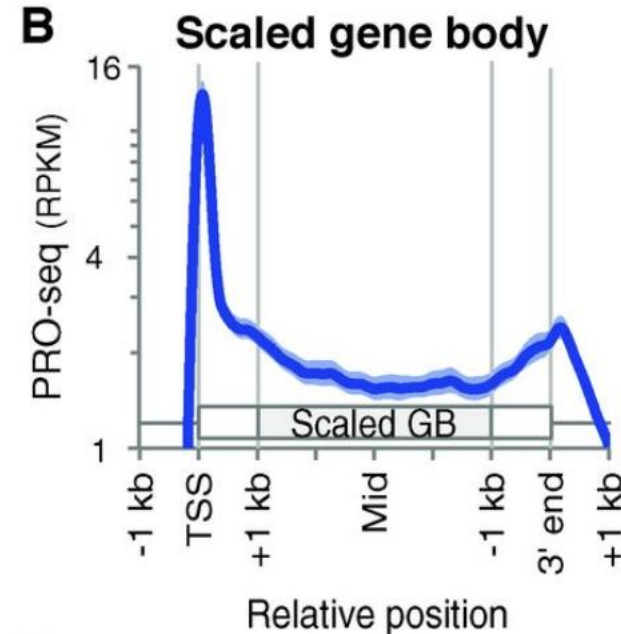
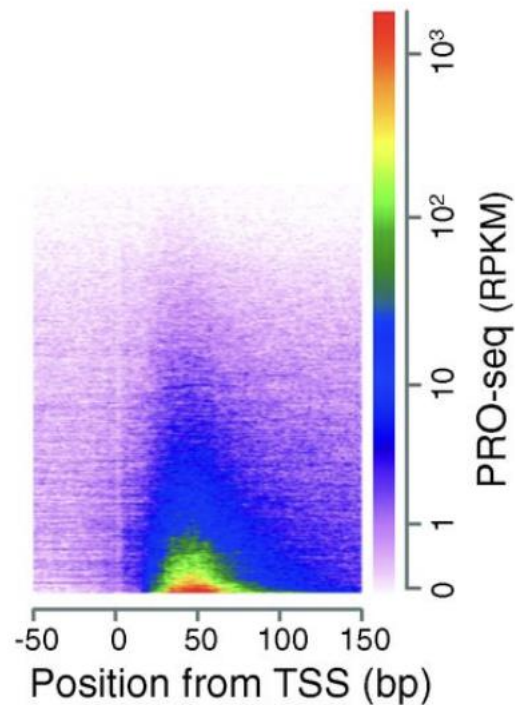
- Maps position of transcriptionally engaged RNA polymerases
- Determines relative activity of transcription sites
- Detects transcription anywhere on the genome
- No prior knowledge of transcription sites is needed

Cons:

- Limited to cell cultures and other artificial systems, due to the requirement for incubation in the presence of labeled nucleotides
- Resolution is only 30–100 nt

Mapping active transcription

PRO-Seq - Precision nuclear run-on sequencing



Mapping active transcription

PRO-Seq - Precision nuclear run-on sequencing



- Single nucleotide resolution of engaged and active RNAP across the genome
- Analysis of polymerase pausing and productive elongation
- Sensitive measurement of differential gene expression
- Identification of enhancers

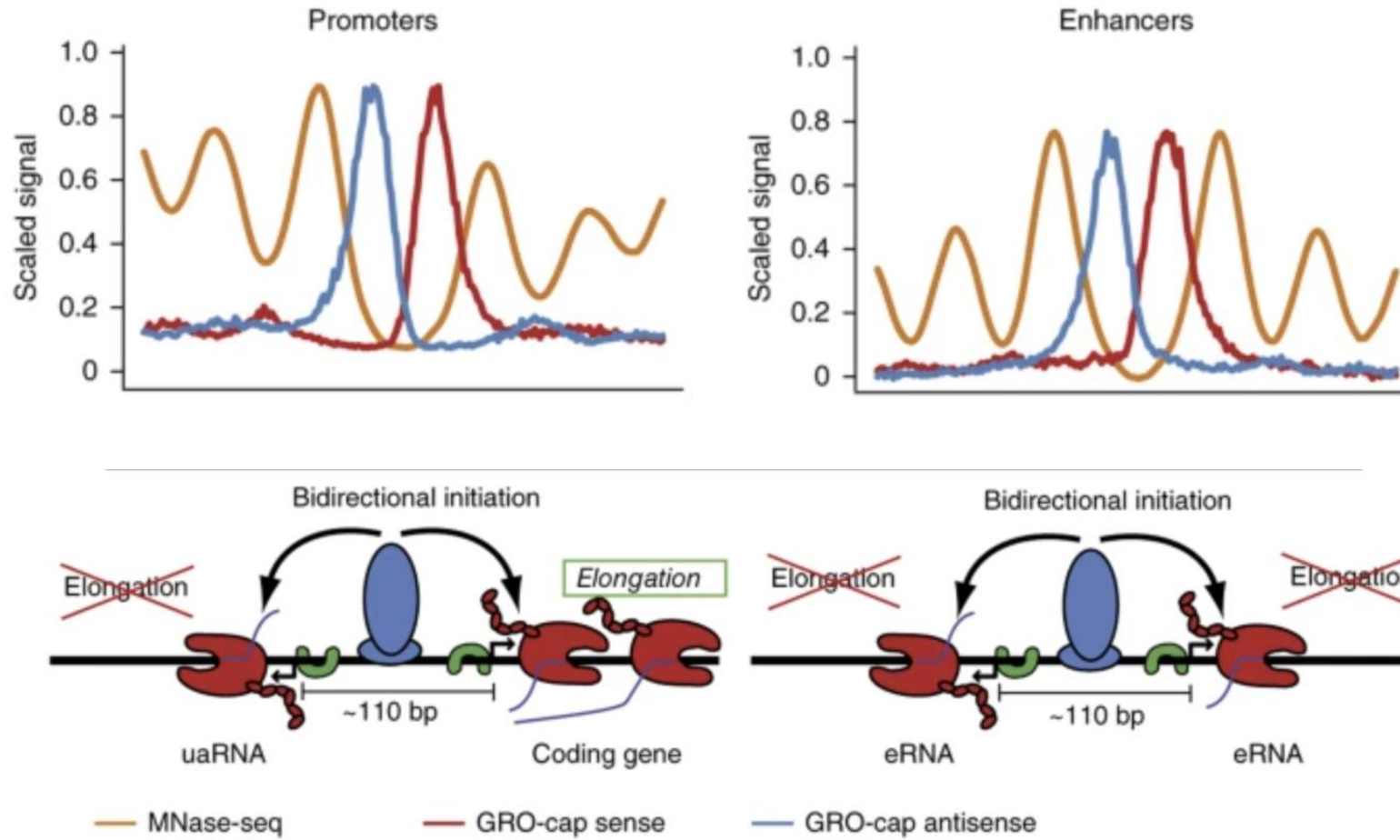
Illumina

Kawak, et al. 2013. Science

Mahat, et al. 2016. Nat Prot

Mapping active transcription

Promoters and enhancers share similar transcription initiation mechanisms



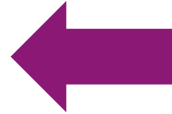
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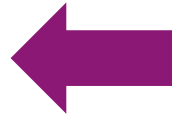


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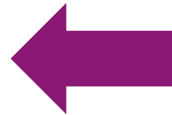
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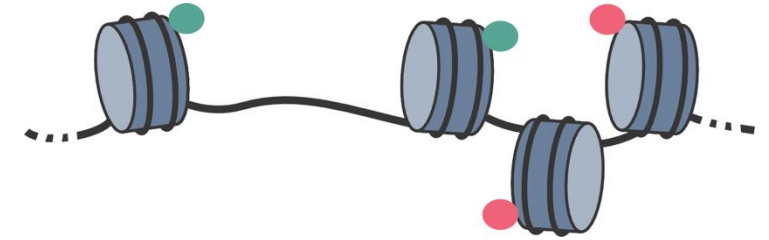
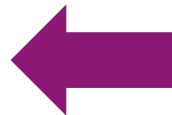
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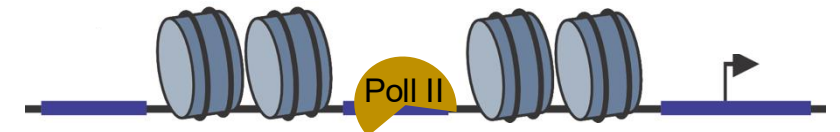
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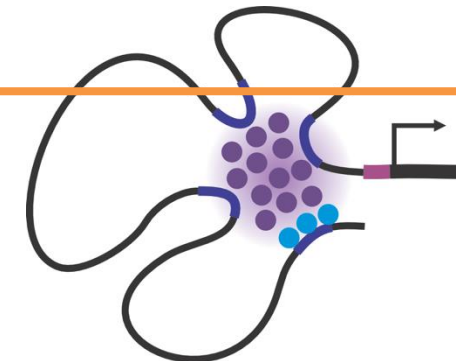
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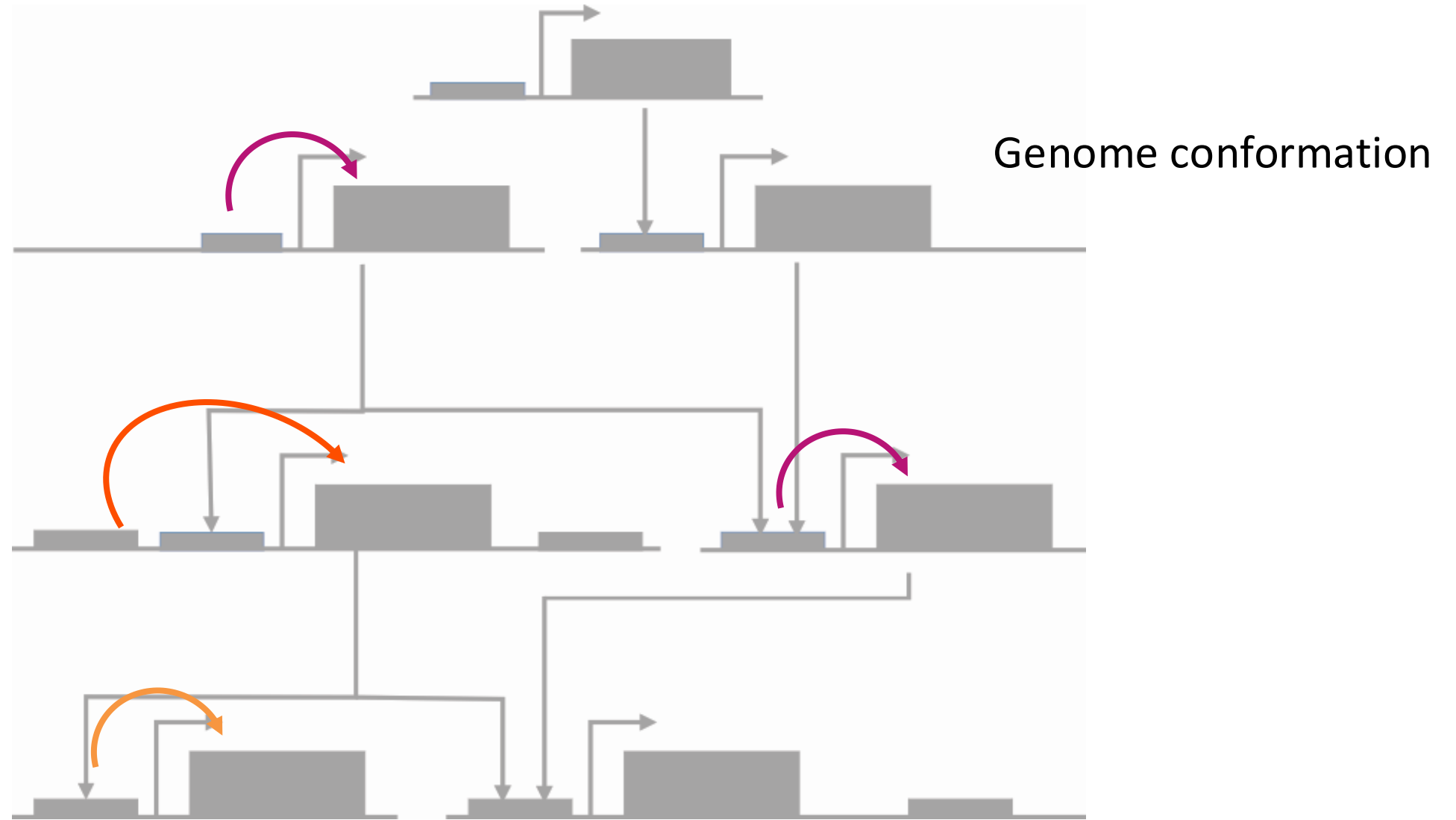


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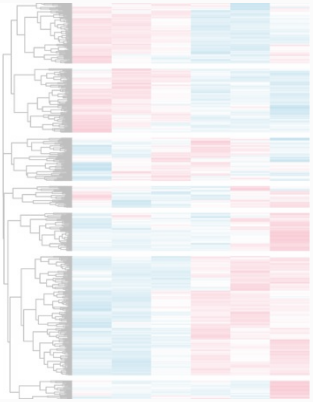
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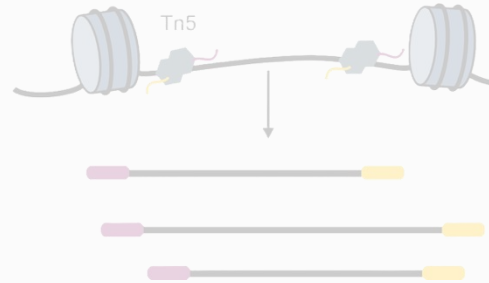
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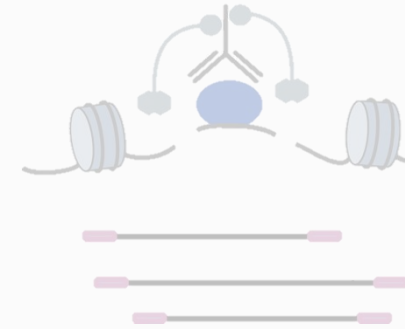
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EPIGENOMIC REGULATION



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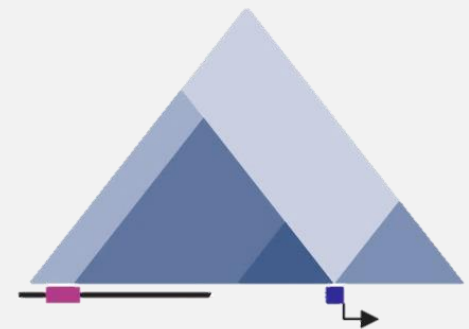
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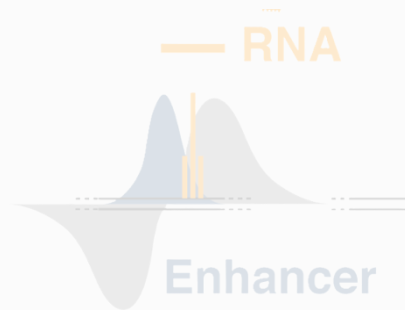
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3D genome organization - HiC



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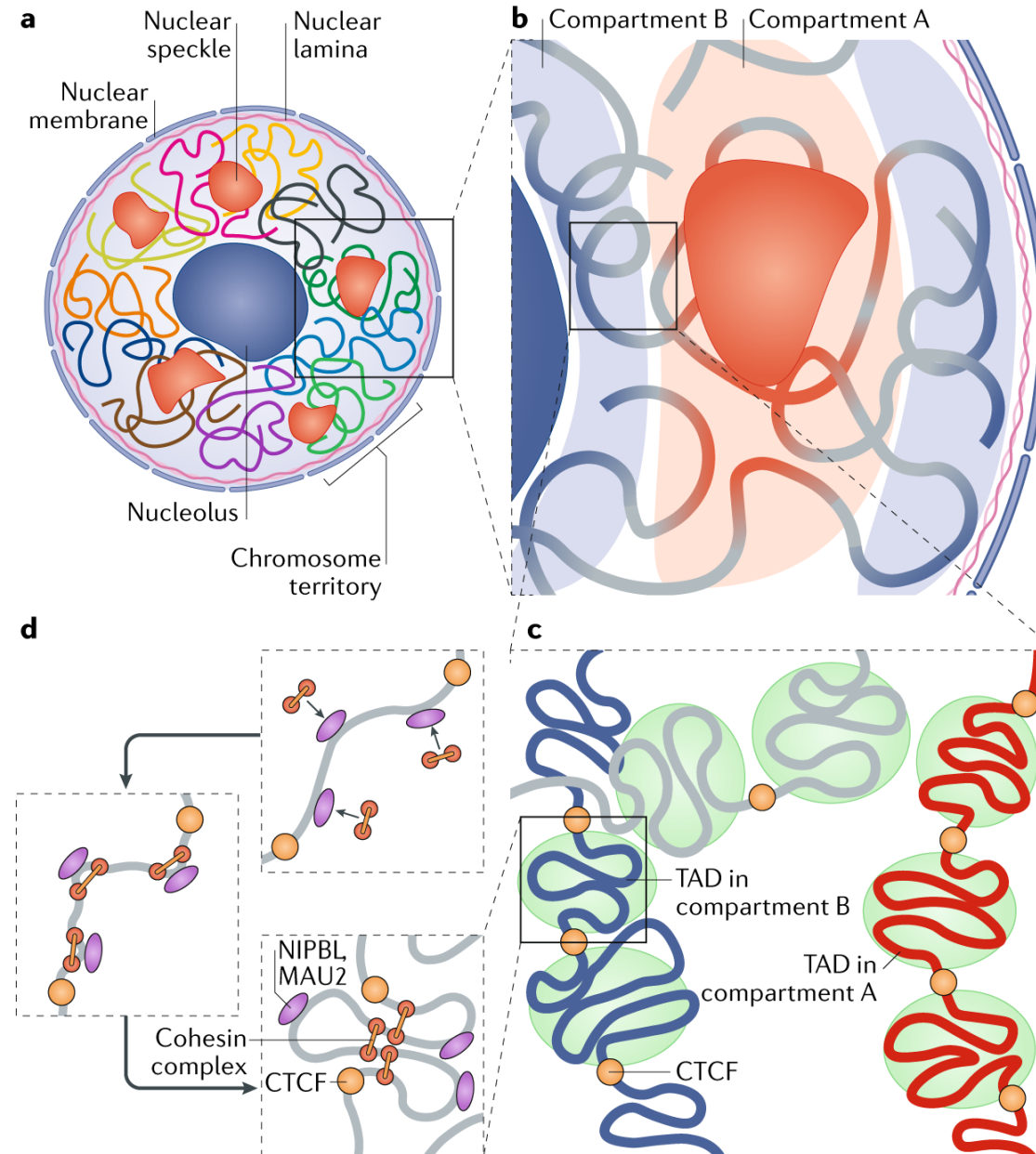


SINGLE-CELL TECHNOLOGIES

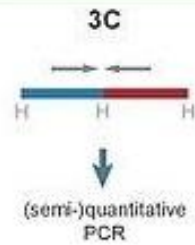
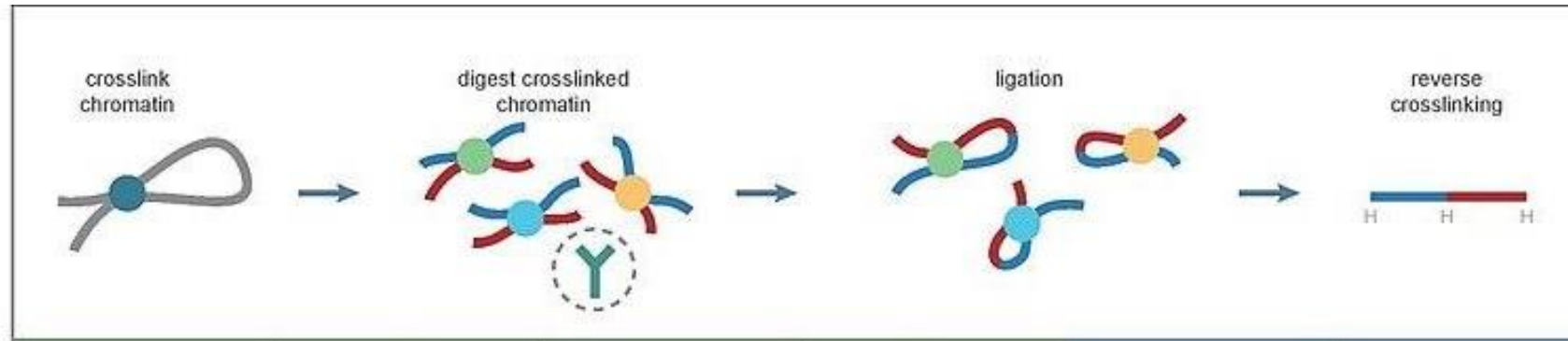
Transcriptomics / Multiome



The hierarchical organization of the 3D genome

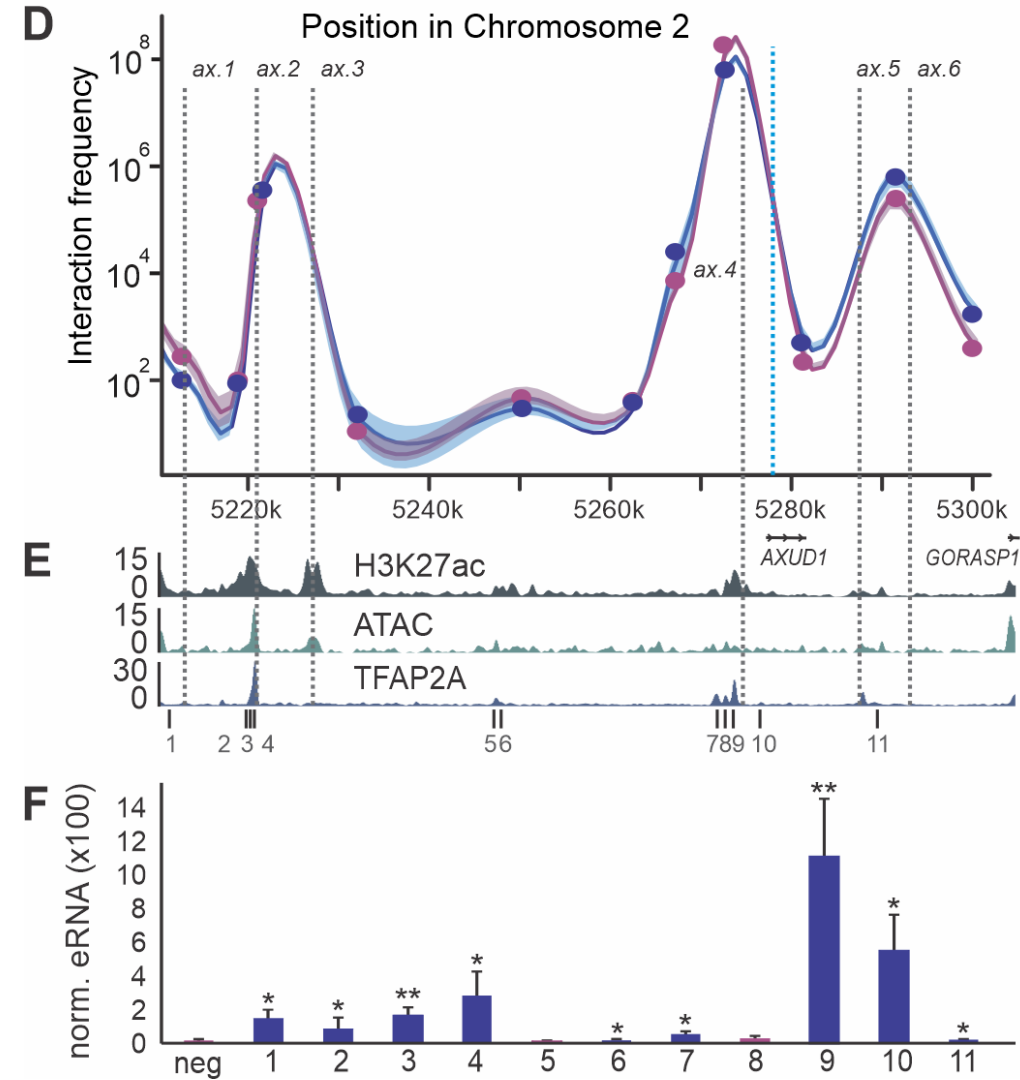
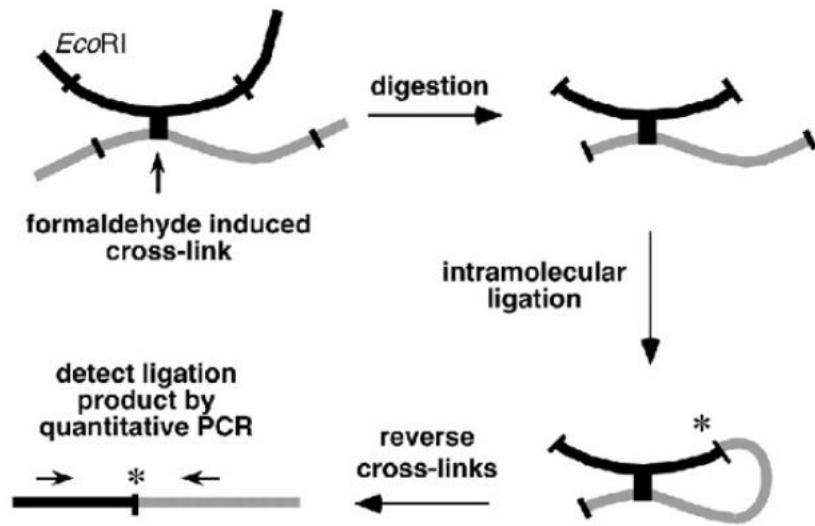


Chromatin Conformation Capture Techniques

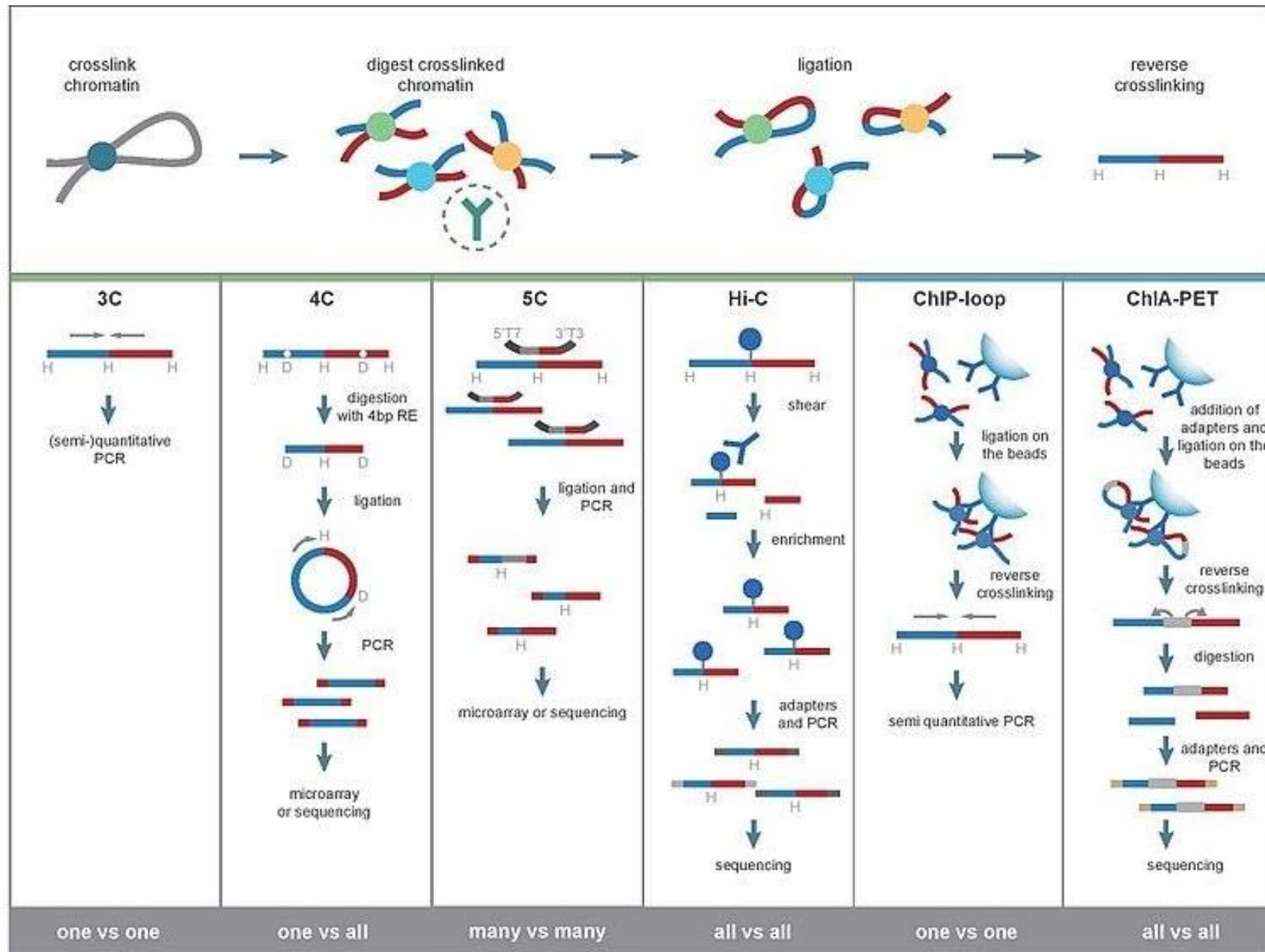


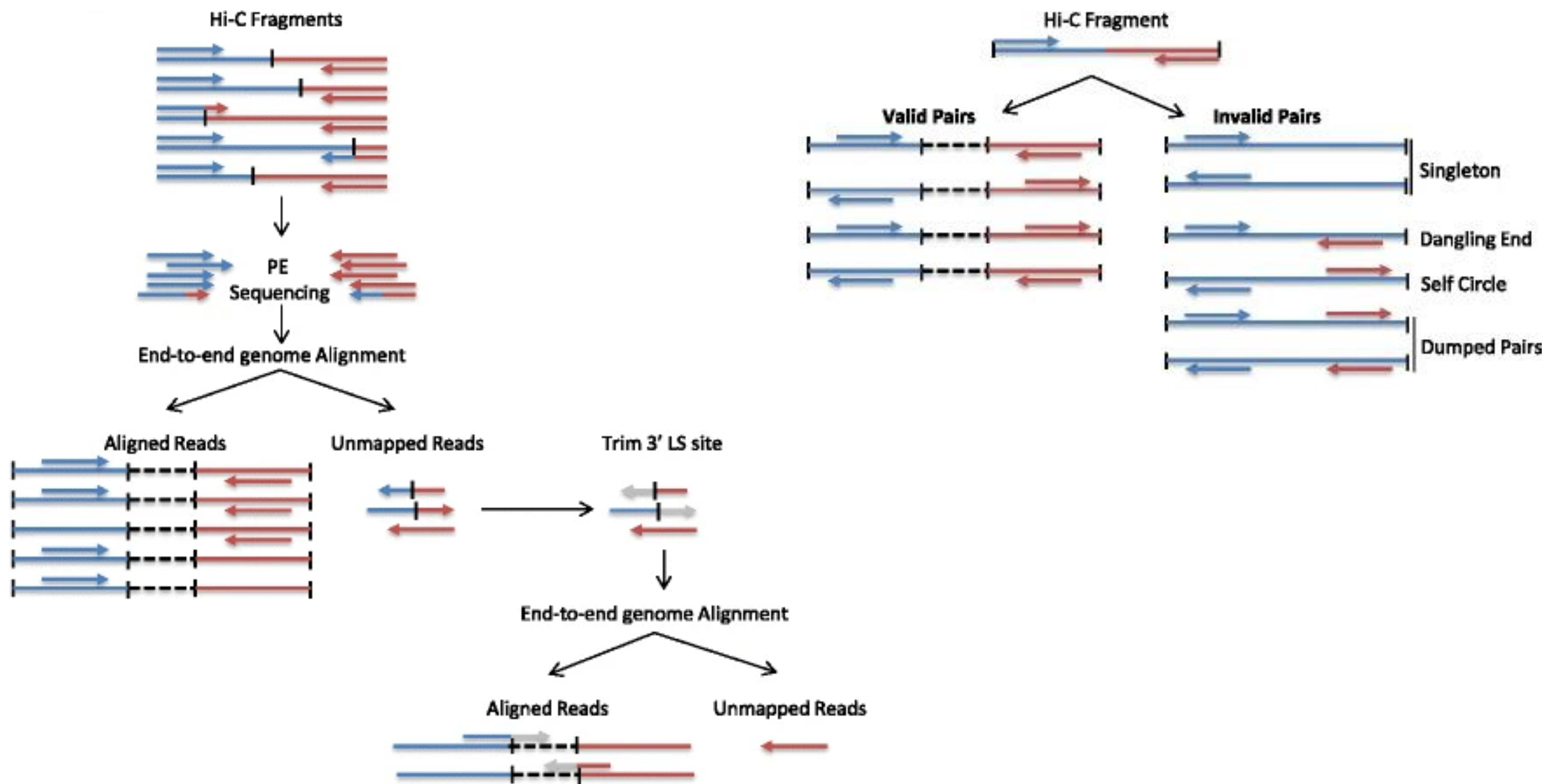
one vs one

Chromatin Conformation Capture Techniques



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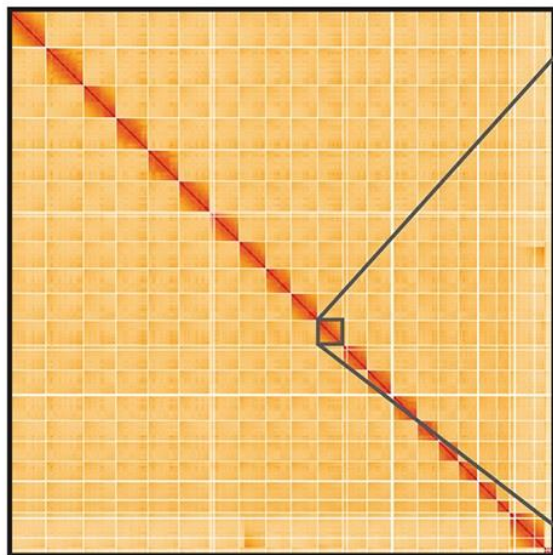




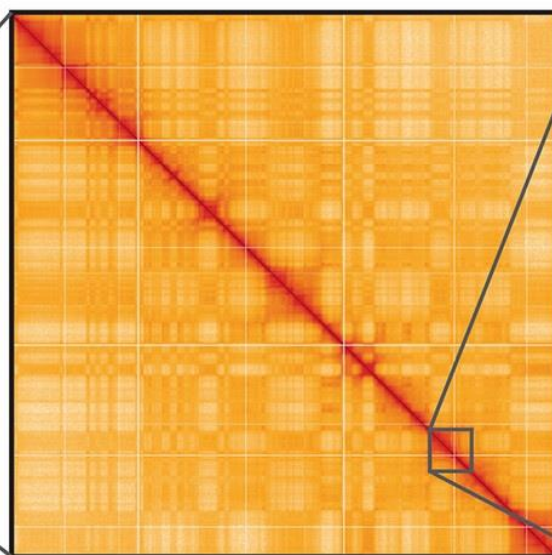
Hi-C

Bonev et al. (2017)

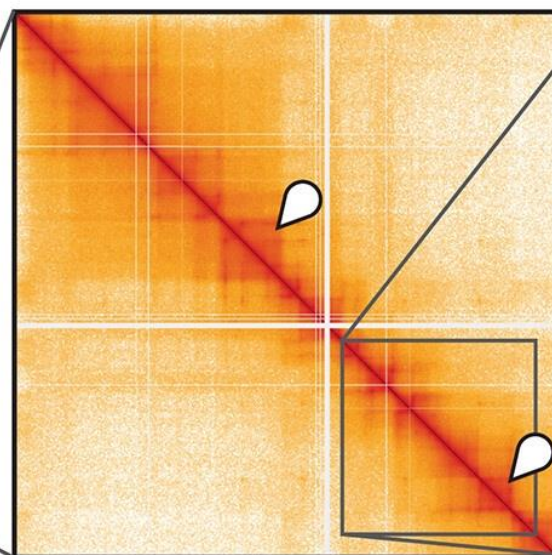
Unique reads = 3.33B

Genome**Compartments**

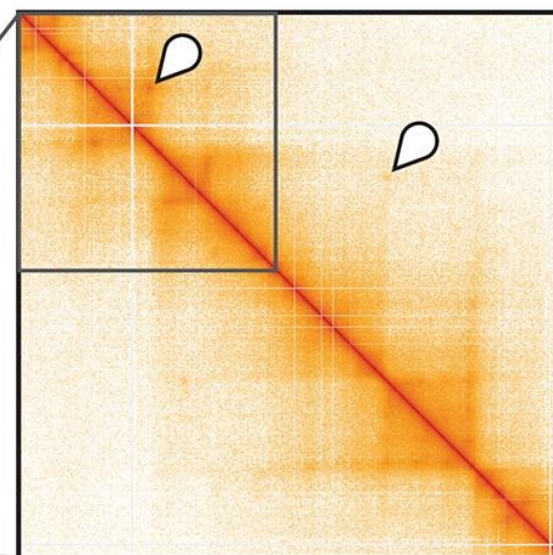
Chr11:35-95M

**TADs**

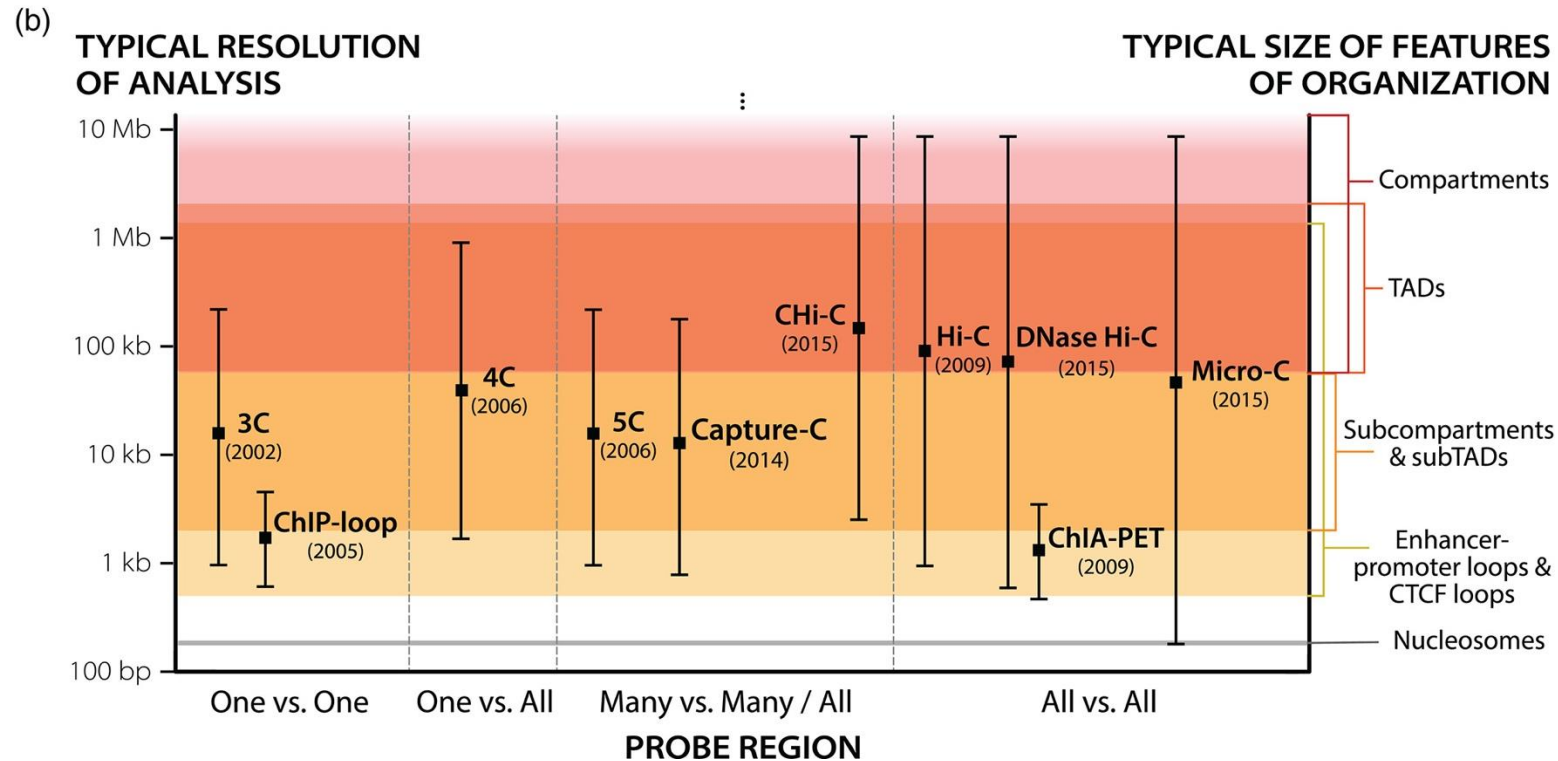
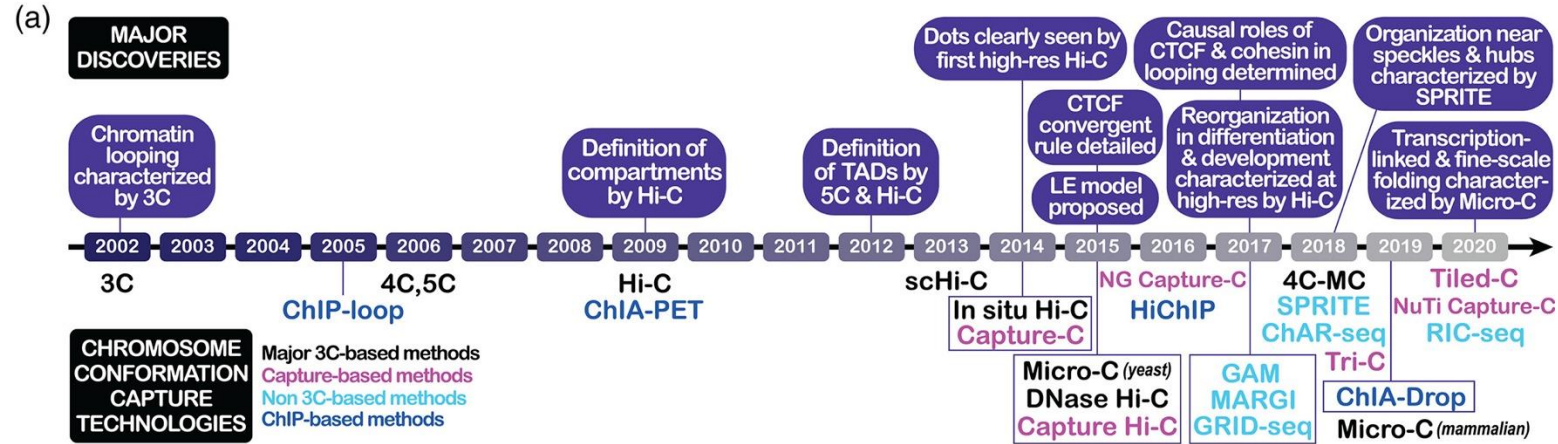
Chr11:80.6-85.4M

**Loops**

Chr11:83.4-85.1M



Chromatin Conformation Capture Techniques



HiChip overview

1. Crosslinking



2. Restriction enzyme cutting and biotin fill-in



3. Proximity ligation



4. Sonication and immunoprecipitation (H3K27ac)



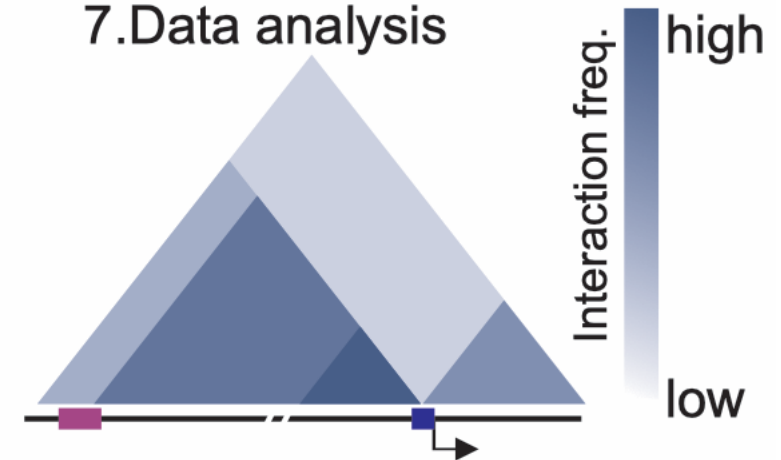
5. DNA purification and biotin enrichment

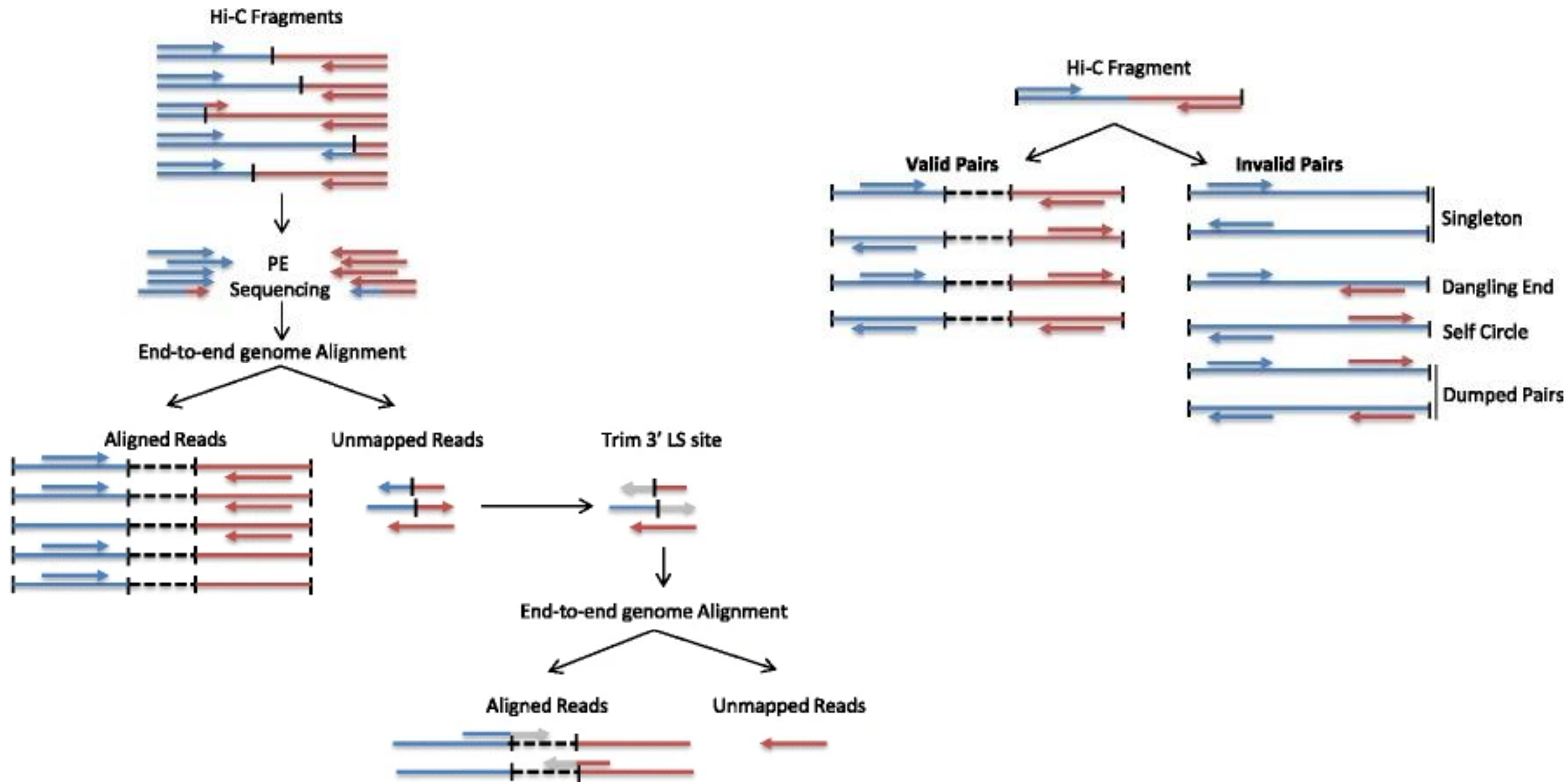


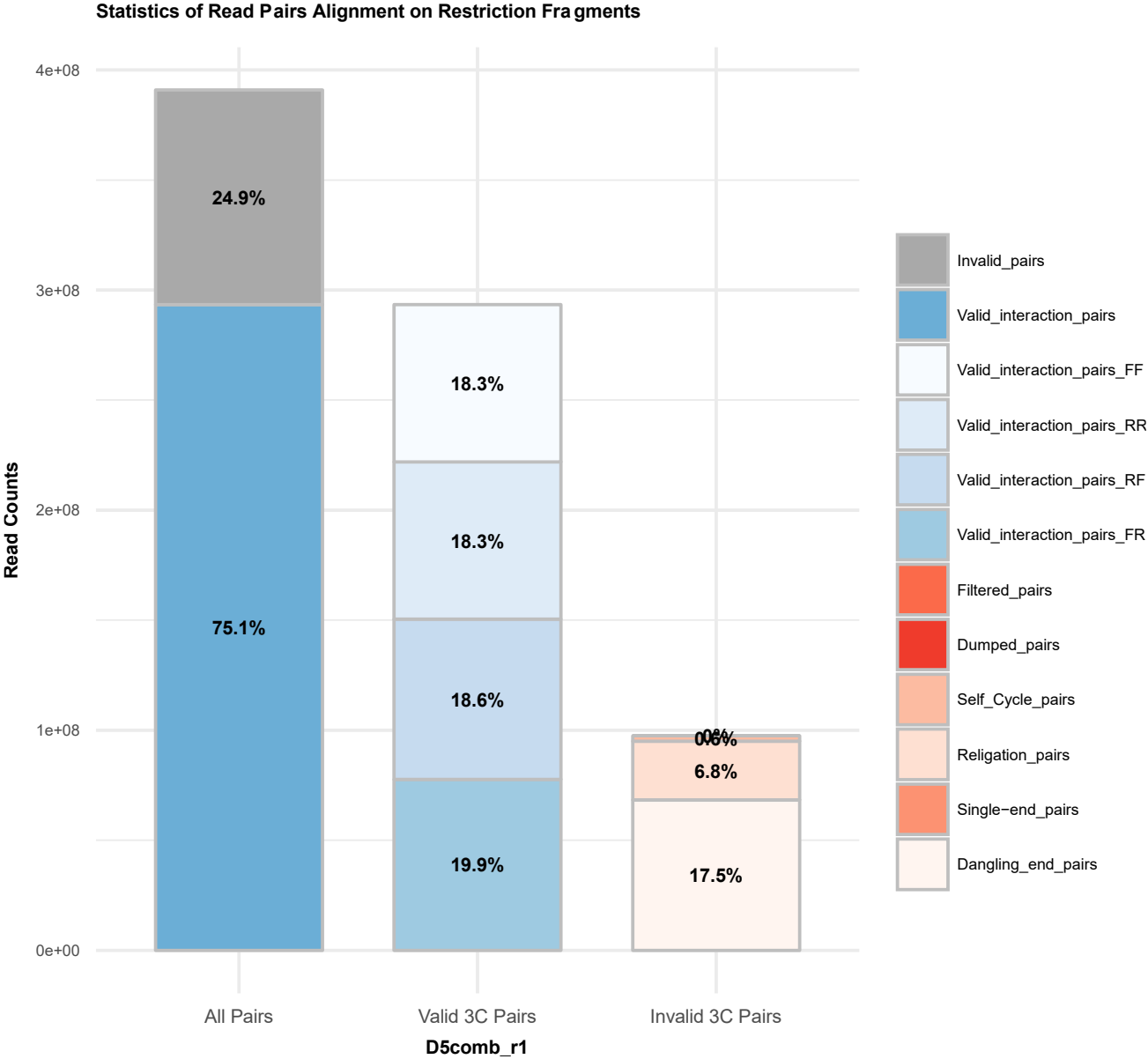
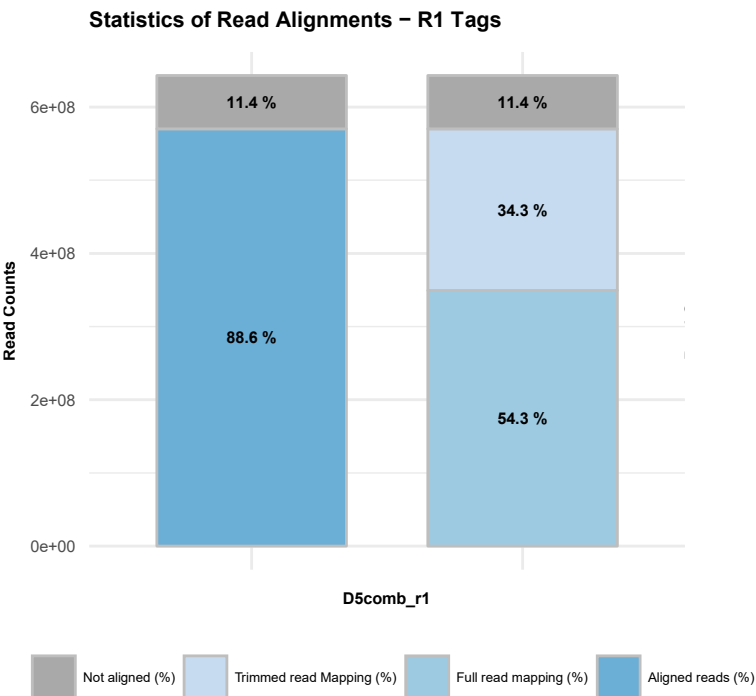
6. Pair-end sequencing

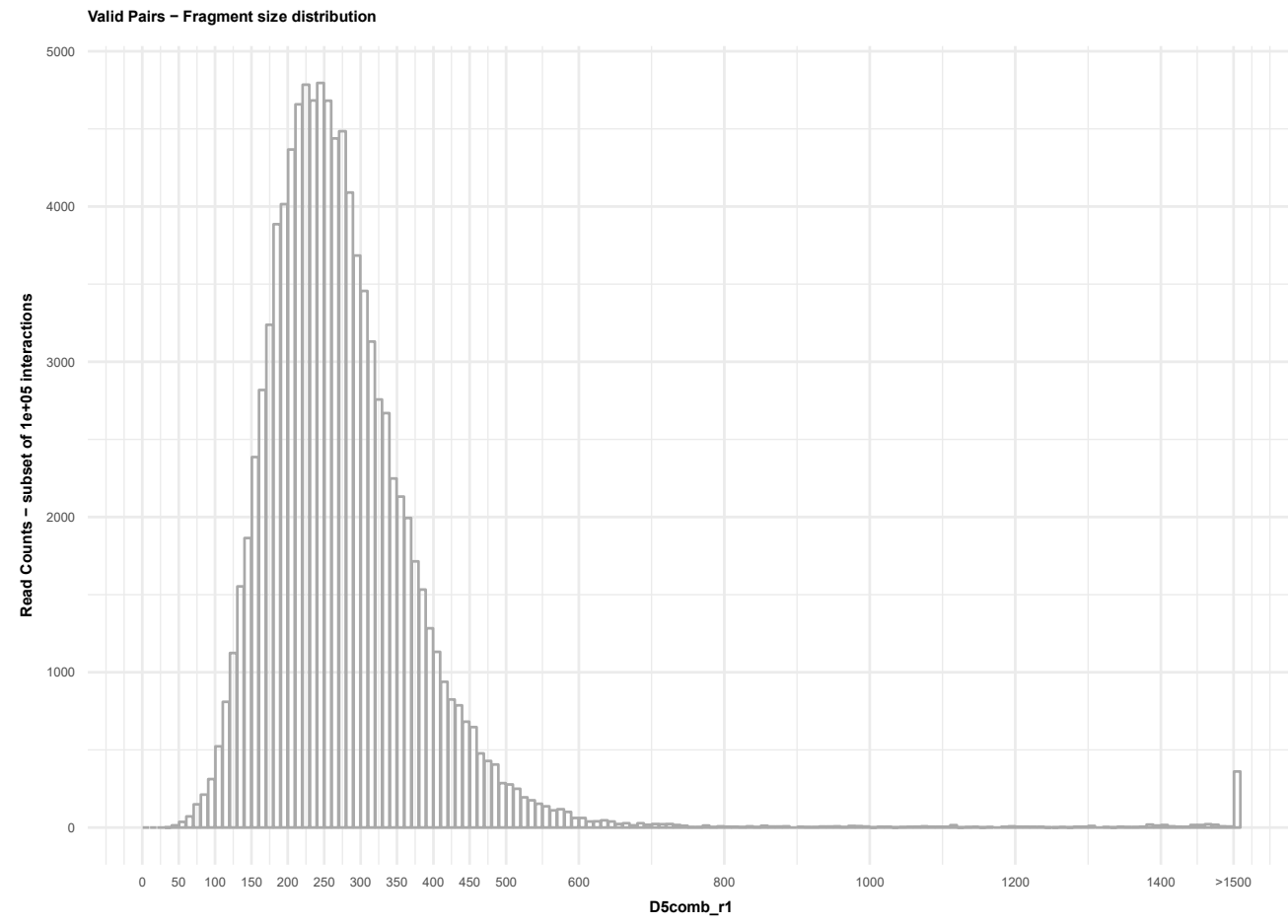
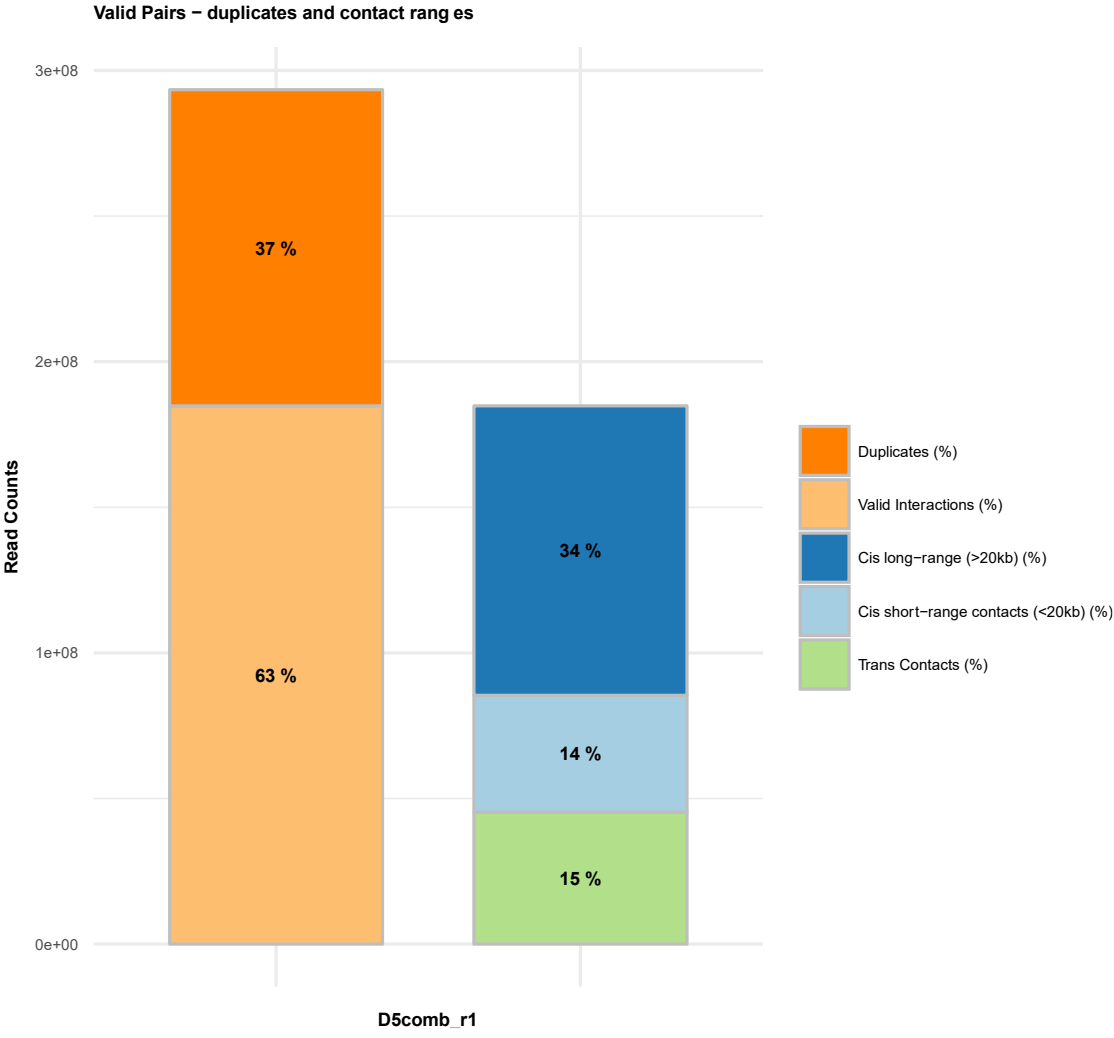


7. Data analysis

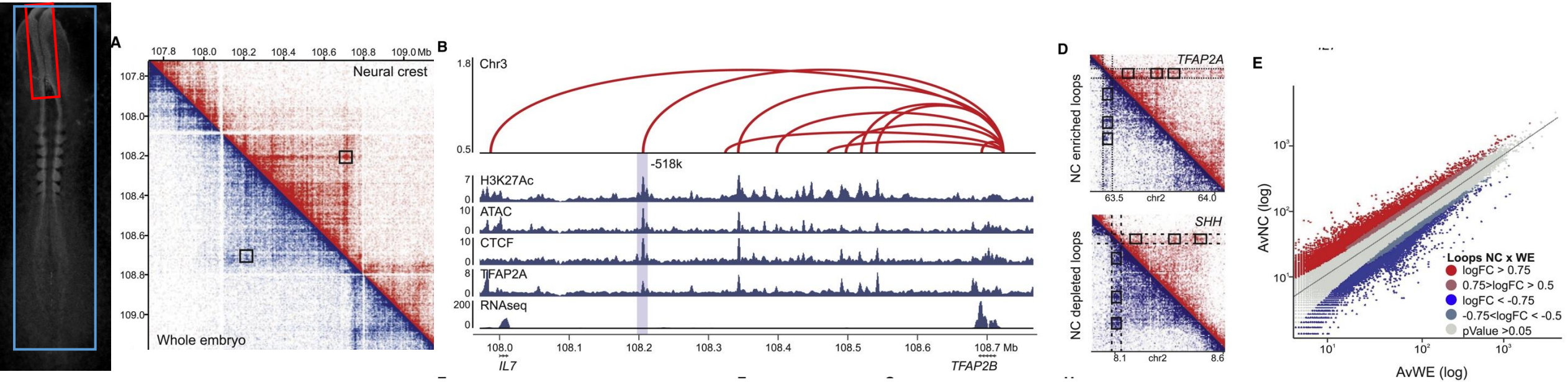






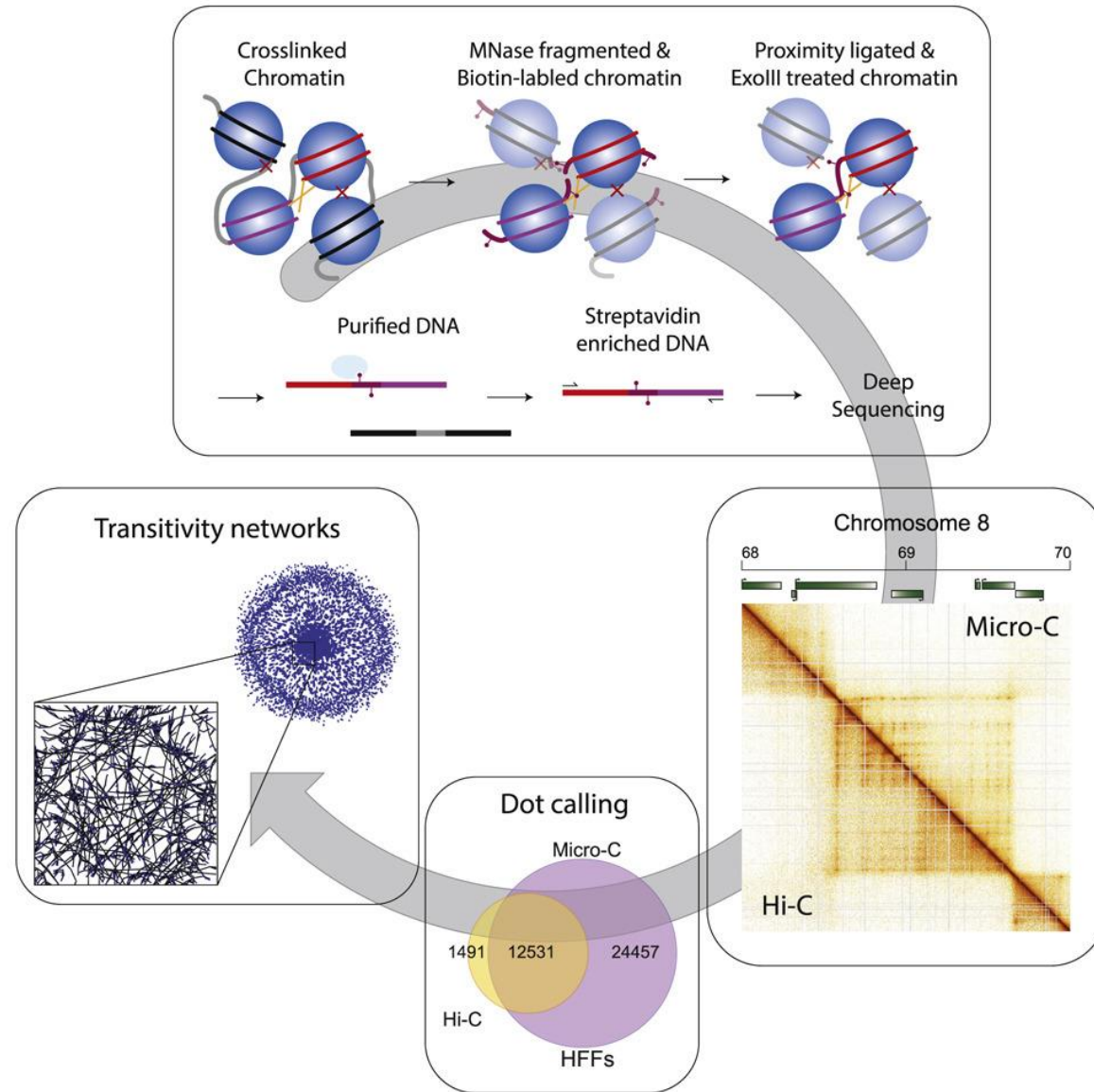


Mapping of tissue-specific enhancer-promoter contacts



Ultrastructural Details of Chromosome Architecture

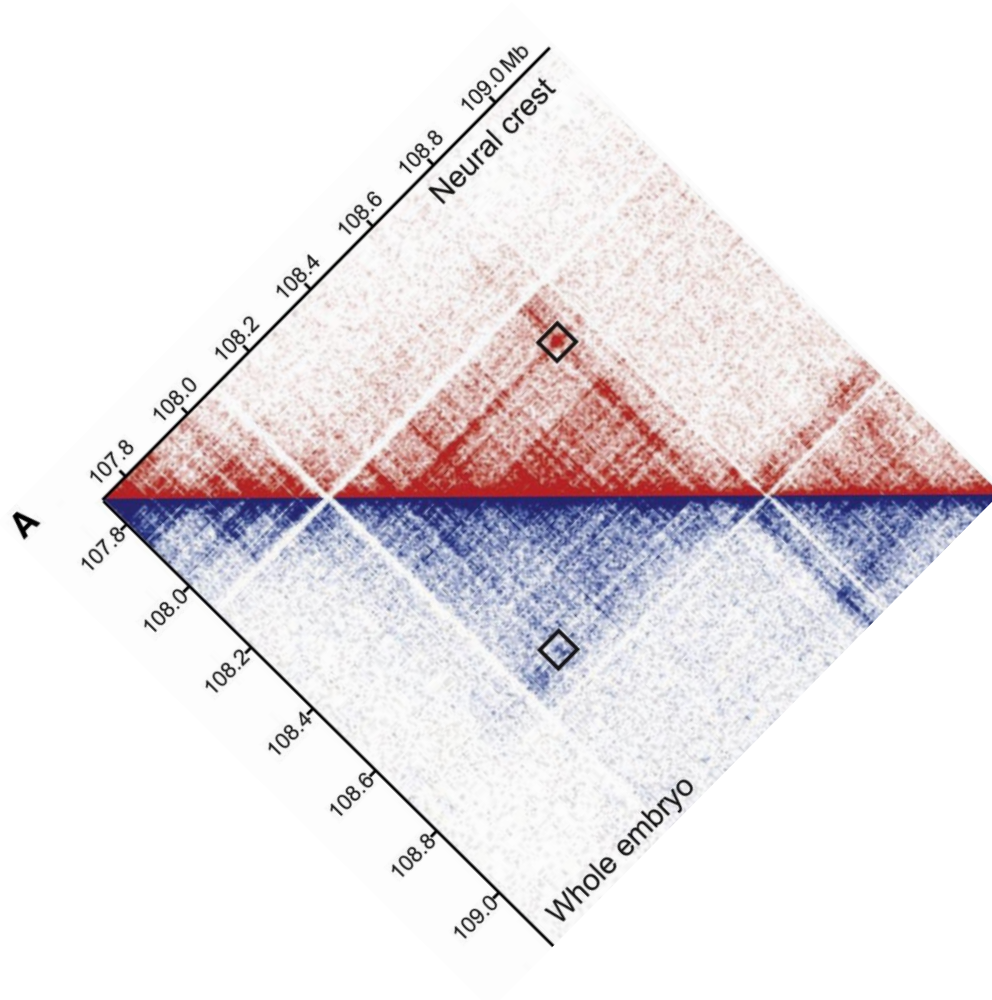
MicroC



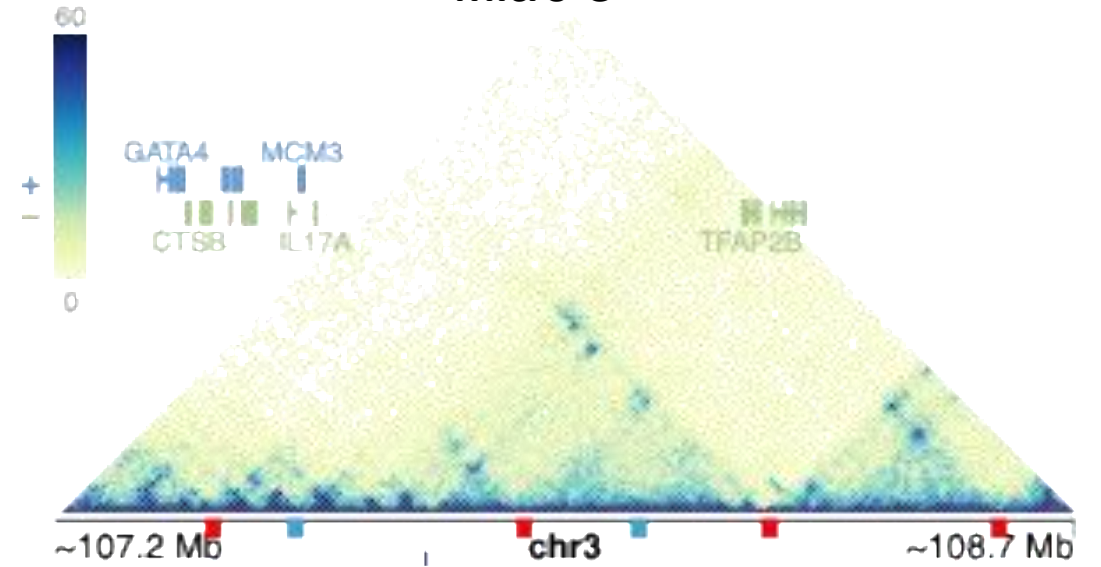
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MicroC

H3K27ac - HiChIP



Micro-C

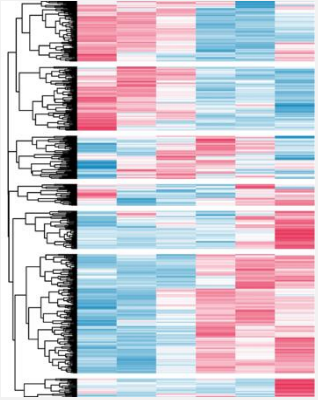


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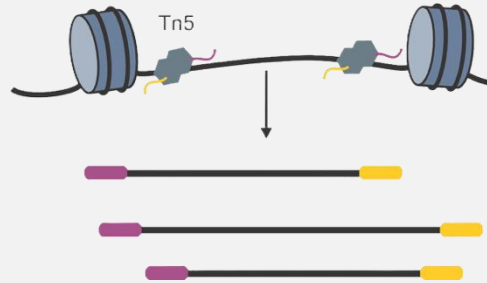
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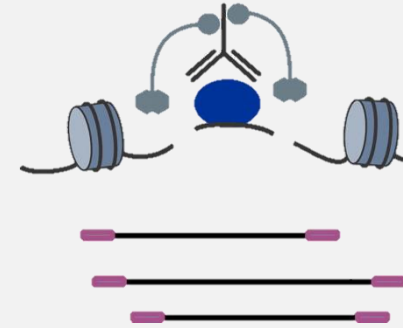
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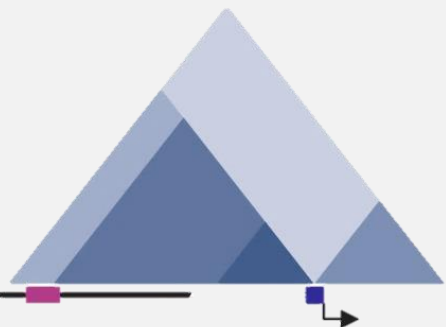
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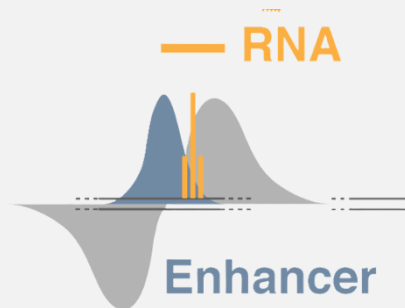
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